

Tutorial on using the Open Food Facts API

Welcome to this tutorial on basic usage of Open Food Facts API.

First, be sure to see our [Introduction to the API](#).

Scan A Product To Get Nutri-score

This basic tutorial shows you can get the Nutri-score of a product, for instance, to display it in a mobile app after scanning the product barcode. Let's use [Nutella Ferrero](#) as the product example for this tutorial.

To get a product nutriscore, send a request to the [Get A Product By Barcode](#) endpoint.

Authentication

Usually, no authentication is required to query Get A Product Nutri-score. However, there is a basic auth to avoid content indexation in the staging environment (which is used throughout this tutorial). For more details, visit the [Open Food Facts API Environment](#).

Describing the Get Request

Make a `GET` request to the `Get A Product By Barcode` endpoint.

```
https://world.openfoodfacts.net/api/v2/product/{barcode}
```

The `{barcode}` is the barcode number of the product you are trying to get. The barcode for **Nutella Ferrero** is **3017624010701**. Then the request path to get product data for **Nutella Ferrero** will look like this:

```
https://world.openfoodfacts.net/api/v2/product/3017624010701
```

The response returns every data about Nutella Ferrero on the database. To get the nutriscore, we need to limit the response by specifying the nutriscore field, which is the

```
nutrition_grades and product_name . {/ /}
```

Query Parameters

To limit the response of the Get A Product By Barcode response, use query parameters to specify the product fields to be returned. In this example, you need one query parameter called `field` with the value `product_name,nutrition_grades`.

The request path will now look like this:

```
https://world.openfoodfacts.net/api/v2/product/3017624010701?
fields=product_name,nutriscore_data
```

Nutri-Score Response

The response returned contains an object of the `code`, `product`, `status_verbose`, and `status`. The `product` object contains the fields specified in the query: the `product_name` and the `nutrition_grades`. The status also states if the product was found or not.

```
{
  "code": "3017624010701",
  "product": {
    "nutrition_grades": "e",
    "product_name": "Nutella"
  },
  "status": 1,
  "status_verbose": "product found"
}
```

Nutri-Score Computation

If you would like to be able to show how the score is computed, add some extra fields like `nutriscore_data` and `nutriments`.

The request path to get the Nutri-Score computation for Nutella-Ferrero will be :

```
https://world.openfoodfacts.net/api/v2/product/3017624010701?
fields=product_name,nutriscore_data,nutriments,nutrition_grades
```

The `product` object in the response now contains the extra fields to show how the nutriscore was computed.

```
{
  "code": "3017624010701",
  "product": {
    "nutriments": {
      "carbohydrates": 57.5,
      "carbohydrates_100g": 57.5,
      "carbohydrates_unit": "g",
      "carbohydrates_value": 57.5,
      "energy": 2255,
      "energy-kcal": 539,

```

```

        "energy-kcal_100g": 539,
        "energy-kcal_unit": "kcal",
        ...,
        ...,
        "sugars": 56.3,
        "sugars_100g": 56.3,
        "sugars_unit": "g",
        "sugars_value": 56.3
    },
    "nutriscore_data": {
        "energy": 2255,
        "energy_points": 6,
        "energy_value": 2255,
        ...,
        ...,
        "sugars_points": 10,
        "sugars_value": 56.3
    },
    "nutrition_grades": "e",
    "product_name": "Nutella"
},
"status": 1,
"status_verbose": "product found"
}

```

For more details, see the reference documentation for [Get A Product By Barcode](#).

```
{/ }
```

Completing products to get the Nutri-Score

Products without a Nutri-Score

When these fields are missing in a nutriscore computation response, it signifies that the product does not have a Nutri-Score computation due to some missing nutrition data. Let's look at the [100% Real Orange Juice](#). If the product nutrition data is missing some fields, you can volunteer and contribute to it by getting the missing tags and writing to the OFF API to add them.

```
{/ }
```

To know the missing tags, check the `misc-tags` field from the product response.

```
https://world.openfoodfacts.net/api/v2/product/0180411000803/100-real-orange-juice?fields=misc_tags
```

The response shows the missing fields and category needed to compute the Nutri-Score.

```

{
  "code": "0180411000803",
  "product": {

```

```

    "misc_tags": [
      "en:nutriscore-not-computed",
      "en:nutriscore-missing-category",
      "en:nutrition-not-enough-data-to-compute-nutrition-score",
      "en:nutriscore-missing-nutrition-data",
      "en:nutriscore-missing-nutrition-data-sodium",
      "en:ecoscore-extended-data-not-computed",
      "en:ecoscore-not-computed",
      "en:main-countries-new-product"
    ]
  },
  "status": 1,
  "status_verbose": "product found"
}

```

The sample response above for 100% Real Orange Juice `misc_tags` shows that the Nutri-Score is missing category (`en:nutriscore-missing-category`) and sodium (salt) (`en:nutriscore-missing-nutrition-data-sodium`). Now you can write to the OFF API to provide these nutriment data (if you have them) so that the Nutri-Score can be computed.

Write data to make Nutri-Score computation possible

The WRITE operations in the OFF API require authentication. Therefore you need a valid `user_id` ^[`user_id_not_email`] and `password` to write the missing nutriment data to 100% Real Orange Juice.

Sign up on the [Open Food Facts App](#) to get your `user_id` and `password` if you don't have one.

To write data to a product, make a `POST` request to the [Add or Edit A Product](#) endpoint.

```
https://world.openfoodfacts.net/cgi/product_jqm2.pl
```

Add your valid `user_id` and `password` as body parameters to your request for authentication. The `code` (barcode of the product to be added/edited), `user_id`, and `password` are required when adding or editing a product. Then, include other product data to be added in the request body.

To write `sodium` and `category` to 100% Real Orange Juice so that the Nutri-Score can be computed, the request body should contain these fields:

Key	Value	Description
<code>user_id</code>	***	A valid <code>user_id</code>
<code>password</code>	***	A valid password

Key	Value	Description
code	0180411000803	The barcode of the product to be added/edited
nutriment_sodium	0.015	Amount of sodium
nutriment_sodium_unit	g	Unit of sodium relative to the amount
categories	Orange Juice	Category of the Product
Using curl:		

```
curl -XPOST -x POST https://world.openfoodfacts.net/cgi/product_jqm2.pl \
-F user_id=your_user_id -F password=your_password \
-F code=0180411000803 -F nutrient_sodium=0.015 -F nutrient_sodium_unit=g -
-F categories="Orange Juice"
```

If the request is successful, it returns a response that indicates that the fields have been saved.

```
{
  "status_verbose": "fields saved",
  "status": 1
}
```

^[user_id_not_email]: user_id is the username of your account. You must not use your email address.

Read newly computed Nutri-Score

Now, let's check if the Nutri-Score for 100% Real Orange Juice has been computed now that we have provided the missing data. Make a GET request to

`https://world.openfoodfacts.net/api/v2/product/0180411000803?`

`fields=product_name,nutriscore_data,nutriments,nutrition_grades` for Nutri-Score of 100% Real Orange Juice. The response now contains the Nutri-Score computation:

```
{
  "code": "0180411000803",
  "product": {
    "nutriments": {
      "carbohydrates": 11.864406779661,
      .
      .
      .
      "sugars_unit": "g",
```

```

        "sugars_value": 11.864406779661
      },
      "nutriscore_data": {
        "energy": 195,
        "energy_points": 7,
        "energy_value": 195,
        .
        .
        .
        "sugars_value": 11.86
      },
      "nutrition_grades": "c",
      "product_name": "100% Real Orange Juice"
    },
    "status": 1,
    "status_verbose": "product found"
  }
}

```

For more details, see the reference documentation for [Add or Edit A Product](#)

You can also check the reference cheatsheet to know how to add/edit other types of product data.

```
{/ }
```

Search for a Product by Nutri-score

Using the Open Food Facts API, you can filter products based on different criteria. To search for products in the Orange Juice category with a nutrition_grade of `c`, query the [Search for Products endpoint](#).

Note: we describe here the v2 search API, but beware that only [v1 search API](#) supports full text search.

Describing the Search Request

Make a `GET` request to the `Search for Products` endpoint.

```
https://world.openfoodfacts.org/api/v2/search
```

Add the search criteria used to filter the products as query parameters. For Orange Juice with a nutrition_grade of `c`, add query parameters `categories_tags_en` to filter `Orange Juice` while `nutrition_grades_tags` to filter `c`. The response will return all the products in the database with the category `Orange Juice` and nutrition_grade `c`.

```
https://world.openfoodfacts.net/api/v2/search?categories_tags_en=Orange
Juice&nutrition_grades_tags=c
```

To limit the response, add `fields` to the query parameters to specify the fields to be returned in each product object response. For this tutorial, limit the response to `code`, `product_name`, `nutrition_grades`, and `categories_tags_en`.

```
https://world.openfoodfacts.net/api/v2/search?categories_tags_en=Orange Juice&nutrition_grades=c&fields=code,nutrition_grades,categories_tags_en
```

The response returns all products that belong to the Orange Juice category, with the `nutrition_grade` "c" and limits each product object response to only the specified fields. It also returns the count(total number) of products that match the search criteria.

```
{
  "count": 1629,
  "page": 1,
  "page_count": 24,
  "page_size": 24,
  "products": [
    {
      "categories_tags_en": [
        "Plant-based foods and beverages",
        "Beverages",
        "Plant-based beverages",
        "Fruit-based beverages",
        "Juices and nectars",
        "Fruit juices",
        "Concentrated fruit juices",
        "Orange juices",
        "Concentrated orange juices"
      ],
      "code": "3123340008288",
      "nutrition_grades": "c"
    },
    .
    .
    .
    {
      "categories_tags_en": [
        "Plant-based foods and beverages",
        "Beverages",
        "Plant-based beverages",
        "Fruit-based beverages",
        "Juices and nectars",
        "Fruit juices",
        "Non-Alcoholic beverages",
        "Orange juices",
        "Squeezed juices",
        "Squeezed orange juices"
      ],
      "code": "3608580844136",
      "nutrition_grades": "c"
    }
  ],
}
```

```
}
  "skip": 0
}
```

Sorting Search Response

You can proceed to also sort the search response by different fields, for example, sort by the product that was modified last or even by the `product_name`. Now, let's sort the products with Orange Juice and a `nutrition_grade` of "c" by when they were last modified. To sort the search response, add the `sort_by` with value `last_modified_t` as a query parameter to the request.

```
https://world.openfoodfacts.net/api/v2/search?
nutrition_grades_tags=c&fields=code,nutrition_grades,categories_tags_en&categories_tags_en=Orange Juice&sort_by=last_modified_t
```

The date that each product was last modified is now used to order the product response.

```
{
  "count": 1629,
  "page": 1,
  "page_count": 24,
  "page_size": 24,
  "products": [
    {
      "categories_tags_en": [
        "Plant-based foods and beverages",
        "Beverages",
        "Plant-based beverages",
        "Fruit-based beverages",
        "Juices and nectars",
        "Fruit juices",
        "Orange juices"
      ],
      "code": "3800014268048",
      "nutrition_grades": "c"
    },
    ,
    ,
    ,
    {
      "categories_tags_en": [
        "Plant-based foods and beverages",
        "Beverages",
        "Plant-based beverages",
        "Fruit-based beverages",
        "Juices and nectars",
        "Fruit juices",
        "Orange juices",
        "Squeezed juices",
        "Squeezed orange juices"
      ],
      "code": "4056489641018",
      "nutrition_grades": "c"
    }
  ]
}
```



```
    ],  
    "skip": 0  
  }
```

To see other examples of sorting a search response, see the reference documentation for [Search for Products](#).